

Examiner
only

7. Describe how the reactions of lithium, sodium and potassium with water can be used to show the trend in reactivity in Group 1. Explain this trend in terms of the electronic structures of the elements. [6 QER]

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END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

1	H	Hydrogen	1
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7	Li	Lithium	3
9	Be	Beryllium	4
23	Na	Sodium	11
24	Mg	Magnesium	12
39	K	Potassium	19
40	Ca	Calcium	20
86	Rb	Rubidium	37
88	Sr	Strontium	38
133	Cs	Caesium	55
223	Fr	Francium	87
137	Ba	Barium	56
226	Ra	Radium	88
227	Ac	Actinium	89

11	B	Boron	5
12	C	Carbon	6
14	N	Nitrogen	7
16	O	Oxygen	8
19	F	Fluorine	9
20	Ne	Neon	10
27	Al	Aluminium	13
28	Si	Silicon	14
31	P	Phosphorus	15
32	S	Sulfur	16
35.5	Cl	Chlorine	17
40	Ar	Argon	18
70	Ga	Gallium	31
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
128	Te	Tellurium	52
127	I	Iodine	53
131	Xe	Xenon	54
204	Tl	Thallium	81
207	Pb	Lead	82
209	Bi	Bismuth	83
210	Po	Polonium	84
210	At	Astatine	85
222	Rn	Radon	86

59	Co	Cobalt	27
56	Fe	Iron	26
55	Mn	Manganese	25
52	Cr	Chromium	24
51	V	Vanadium	23
48	Ti	Titanium	22
45	Sc	Scandium	21
89	Y	Yttrium	39
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
99	Tc	Technetium	43
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
108	Ag	Silver	47
112	Cd	Cadmium	48
63.5	Cu	Copper	29
65	Zn	Zinc	30

Key

Ar	Symbol	Name	Z
			relative atomic mass
			atomic number

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430UB0-1



SCIENCE (Double Award)

**Unit 2: CHEMISTRY 1
HIGHER TIER**

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	7	
3.	6	
4.	8	
5.	5	
6.	6	
7.	11	
8.	9	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN183430UB0101

7. (a) (i) A group of students were given three water samples. One was soft water, one was temporary hard water and one was permanent hard water.
- Describe a method that the students could use to find out which is which. Give the expected results. You do **not** need to include the detail required to ensure a fair test. [4]

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- (ii) The samples from part (i) were labelled **X**, **Y** and **Z** and sent to have their ion content measured. The table shows the results.

Ions present	Ion content in water sample (mg/dm ³)		
	X	Y	Z
sodium	29	116	23
potassium	12	15	11
magnesium	31	4	98
calcium	141	2	27
hydrogencarbonate	30	19	219
chloride	17	14	20
sulfate	346	6	27
nitrate	12	15	19

Use this information to find whether sample **Z** is the soft water, the temporary hard water or the permanent hard water. Give a reason for your choice. [2]

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- (b) (i) Permanent hard water can be softened by ion exchange. Explain how ion exchange works. [2]

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- (ii) Explain why the resin in ion exchange stops working after continued use. [1]

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- (c) A litre bottle of mineral water contains 184mg of dissolved calcium sulfate, CaSO_4 . Calculate the number of moles of calcium sulfate present.

Give your answer to **three** significant figures.

1 mg = 0.001 g [2]

$A_r(\text{Ca}) = 40$ $A_r(\text{S}) = 32$ $A_r(\text{O}) = 16$

Moles of calcium sulfate = mol

11



7. (a) (i) Chlorine is a non-metal found in Group 7 of the Periodic Table. When it is bubbled into a solution of potassium iodide there is a colour change from pale green to brown. Explain why this reaction occurs. [2]

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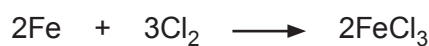
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- (ii) Write the balanced symbol equation for the reaction between chlorine and potassium iodide. [2]

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- (b) The symbol equation for the reaction between iron and chlorine is as follows.



Calculate the mass of chlorine needed to react with 1.32 g of iron. [3]

$$A_r(\text{Fe}) = 56$$

$$A_r(\text{Cl}) = 35.5$$

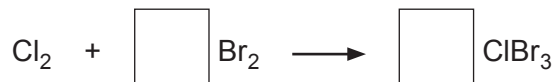
Mass of chlorine = g



- (c) (i) Under certain conditions, Group 7 elements will react with each other to produce new compounds.

When chlorine is reacted with bromine, chlorine tribromide is made.

Balance the symbol equation for this reaction. [1]



- (ii) A chemist calculated that if she reacted 7.00 g of chlorine with an excess of bromine, the theoretical mass of chlorine tribromide produced is 27.55 g.

However, when she carried out the experiment using 7.00 g of chlorine the mass of chlorine tribromide obtained was 21.34 g.

Calculate the percentage yield of chlorine tribromide. [1]

Percentage yield = %



7. The table shows some properties of elements in Period 3 of the Periodic Table.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Appearance	Malleability	Conductivity
Na	98	882	1.00	shiny solid	malleable	good
Mg	650	1091	1.75	shiny solid	malleable	good
Al	660	2470	2.70	shiny solid	malleable	good
Si	1410	3265	2.35	shiny solid	brittle	semiconductor
P	44	281	1.80	red solid	brittle	poor
S	113	444	2.05	yellow solid	brittle	poor
Cl	−101	−34	0.003	green gas	n/a	poor

(a) One of the elements is difficult to classify as a metal or non-metal.

Identify this element and give your reasoning.

[2]

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7. A great deal of scientific research into global warming has happened over the last 30 years. More recently, two new contrasting theories linked to global warming have been widely reported. These are outlined below.

Solar activity

A small number of scientists believe that changes in solar activity (changes in the brightness and warmth of the Sun) is causing global warming. Solar activity occurs on the surface of the Sun when nuclear reactions cause flares to be released, high winds to occur and a release of high energy particles from the Sun.

Global dimming

Some scientists believe that global warming is increasing the Earth’s mean temperature, but that global dimming is preventing it from increasing even more. Global dimming is the reduction in the amount of radiation reaching the Earth’s surface from the Sun. It is thought to be caused by an increase of aerosol particles in the Earth’s atmosphere caused by pollution, dust and smog, as well as volcanic eruptions. These particles absorb solar energy and reflect sunlight back into space.

Figure 1 shows the mean atmospheric temperature, carbon dioxide concentration in the atmosphere and solar activity between 1930 and 2010.

Figure 1

Year	Mean atmospheric temperature (°C)	CO ₂ concentration (ppm)	Solar activity (arbitrary units)
1930	13.6	309	3.9
1940	13.6	311	4.0
1950	13.8	316	4.2
1960	13.9	320	4.2
1970	14.0	331	4.0
1980	14.2	342	3.9
1990	14.3	358	3.7
2000	14.4	370	3.7
2010	14.6	383	3.6

ppm = parts per million

